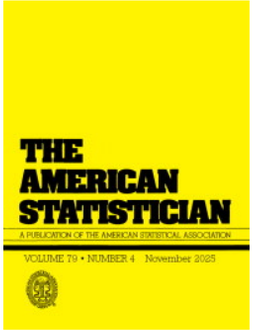

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A Design for Observational Studies in Which Some People Avoid Treatment

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Abstract

A simple but better design is proposed for observational studies in which receipt of treatment is rare in a large subpopulation defined by measured covariates. In such a subpopulation, few responses to treatment are observed, but there are abundant untreated controls. The design is matched in two ratios, a low ratio – perhaps matched pairs – for the subpopulation where treatment is common, and a high ratio – perhaps 1-to-4 – for the subpopulation where treatment is uncommon, retaining this distinction throughout the study, rather than presuming that avoidance of treatment is inconsequential. Theoretical considerations that favor this design are discussed informally with reference to the existing literature on observational studies. These include: increased design sensitivity, possible effect modification and its possible consequences for design sensitivity, counterfactual low risk in relation to design sensitivity, and the degree to which conclusions may be generalized. The design is exemplified by a study with 1212 treatment-control matched pairs and 213 sets matched 1-to-4, so that more than 30% of the $3489 = 2\times1212 + 213 + 4\times213$ study subjects come from the subpopulation that rarely receives treatment. The `R` package `aamatch` contains the unmatched data, reconstructs the matched data, and replicates the analyses.

Keywords: Causal inference; design sensitivity; effect modification; propensity score; sensitivity analysis.

1 Introduction

1.1 Considerations in Research Design

In their article, "Robust designs,"Box and Draper (1975, p. 347) asked "What constitutes a 'good' response surface design?" They answered with 14 considerations, remarking that:

Part of the art of the good practicing statistician is an ability to assess the special needs of a given situation and to choose a design which comes close to meeting them … It would also be helpful to know which criteria are in conflict, and which in harmony.

Notable are: (i) their desire for a "good,"not an "optimal,"design; (ii) the considerations that comprise "good"are at least 14 in number and may be in conflict with each other, (iii) to the extent that good considerations conflict with each other, they also conflict with optimizing

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any one consideration to the exclusion of the others; and (iv) they seek to enhance an "art"and an "ability"rather than to establish a fixed rule.

A design for an observational study, not an experiment, is proposed here. This design is a simple extension of a common design, and a number of considerations speak in its favor; however, I have never seen it used in practice. While I will discuss one design, I am at least as concerned to illustrate some of the reasoning that should go into choosing any design for an observational study. In particular, I want to emphasize weighing conflicting considerations, seeking a design that is not best in terms of any one consideration, but is a better choice than any best design, because it balances important considerations.

1.2 Outline: Design Considerations; Example; Construction

In outline: In a common, useful design for an observational study, each treated individual, $i = 1, \ldots, I$, is matched for covariates to K, controls, where matched pairs with $K = 1$ is most common. It has been known for some time that you can remove more bias from measured covariates, while also using more controls, by letting the number of controls vary depending upon the availability of good controls (Brumberg et al., 2024; Hansen, 2004; Ming and Rosenbaum, 2000; Pimentel et al., 2015; Rosenbaum, 1991).

The proposed design takes a substantial but limited step in this direction: the limitations reflect considerations in §3.1-§3.6 besides adjustment for measured covariates. In this design, most treated people are matched to K controls, but the types of people who seem to avoid treatment are matched to $K' > K$ controls; in the example $K = 1$ and $K' = 4$. It will take a moment's thought in §3 and an example in §4 to see what can, and what cannot, be gained in this way. The easy steps in constructing the design are discussed in §5.

2 Types of People Who are Quite Likely to be Controls

2.1 Notation and Structure

In some observational studies, a large, recognizable group of individuals are rarely exposed to treatment. In §4, people in the US with more income and education are unlikely to smoke cigarettes. If $\mathbf{x}$ contains covariates measured prior to treatment assignment, and if $Z = 1$ for treated individuals and $Z = 0$ for controls, then there is a set $\mathcal{C}$ of values of $\mathbf{x}$ such that $\Pr(\mathbf{x} \in \mathcal{C})$ is moderately large, but the propensity score, $e(\mathbf{x}) = \Pr(Z = 1 | \mathbf{x})$, though not zero, is fairly low for $\mathbf{x} \in \mathcal{C}$. In §4, for $\mathbf{x} \in \mathcal{C}$ about 1 person in 11 is treated, but for $\mathbf{x} \notin \mathcal{C}$ about 1 person in 3 is treated, yet $\approx 40\%$ of the population has $\mathbf{x} \in \mathcal{C}$, with $\mathcal{C}$ is defined by sex, age, education and income. For $\mathbf{x} \in \mathcal{C}$, there are many controls, but comparatively few treated individuals. What is a good study design in this situation?

Each person has two potential outcomes or responses, (r_T, r_C), where r_T is seen under treatment and r_C is seen under control, so the one observed response is $R = Z r_T + (1 - Z) r_C$, and the effect caused by the treatment, $r_T - r_C$, is not observed for any individual; see Neyman (1990) and Rubin (1974). Here, $R = r_C$ is observed for controls $Z = 0$ whether

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$\mathbf{x} \in \mathcal{C}$ or $\mathbf{x} \notin \mathcal{C}$, and R_T is observed for many treated individuals with $Z = 1$ and $\mathbf{x} \notin \mathcal{C}$, but $R = r_T$ is rarely observed for $\mathbf{x} \in \mathcal{C}$, even though $\Pr(\mathbf{x} \in \mathcal{C})$ is fairly large.

Treatment assignment is ignorable given $\mathbf{x}$ (or unconfounded by unmeasured covariates) if

$$0 < \Pr(Z = 1|\mathbf{x}) = \Pr(Z = 1|r_T, r_C, \mathbf{x}) < 1, \quad (1)$$

so $Z \perp\!\!\!\perp (r_T, r_C) \big| \mathbf{x}$, where $e(\mathbf{x}) = \Pr(Z = 1|\mathbf{x})$ is the propensity score and $\zeta = \zeta(r_T, r_C, \mathbf{x}) = \Pr(Z = 1|r_T, r_C, \mathbf{x})$ is the principal unobserved covariate, and assignment is always ignorable given $(\mathbf{x}, \zeta)$; see Rosenbaum and Rubin (2023, §4). When (1) holds, $\mathrm{E}(R|Z = 1, \mathbf{x}) = \mathrm{E}(r_T|Z = 1, \mathbf{x}) = \mathrm{E}(r_T|\mathbf{x})$ and $\mathrm{E}(R|Z = 0, \mathbf{x}) = \mathrm{E}(r_C|\mathbf{x})$, so various causal effects, including $\mathrm{E}(r_T - r_C|\mathbf{x})$, $\mathrm{E}(r_T - r_C)$, and $\mathrm{E}(r_T - r_C|Z = 1)$, may be consistently estimated from the distributions of observable data. Central to every observational study is the problematic nature of (1) in the absence of randomization.

If a model predicts R from $(Z, \mathbf{x})$, then inference about $\mathrm{E}(R|Z = 1, \mathbf{x}) = \mathrm{E}(r_T|Z = 1, \mathbf{x})$ for $\mathbf{x} \in \mathcal{C}$ is largely an extrapolation from three sources: (i) substantial data about $\mathrm{E}(r_T|Z = 1, \mathbf{x})$ for $\mathbf{x} \notin \mathcal{C}$, (ii) substantial data about $\mathrm{E}(R|Z = 0, \mathbf{x}) = \mathrm{E}(r_C|Z = 0, \mathbf{x})$ for most or all $\mathbf{x}$, and (iii) limited information about $\mathrm{E}(r_T|Z = 1, \mathbf{x})$ for $\mathbf{x} \in \mathcal{C}$. This is true whether or not (1) holds. A simple way to define $\mathcal{C}$ is $\mathcal{C} = \{\mathbf{x} : e(\mathbf{x}) \le \kappa\}$ for some κ.

Pimentel et al. (2015) call $\phi(\mathbf{x}) = \{1 - e(\mathbf{x})\} / e(\mathbf{x})$ the "entire number," noting that in an infinite population, we expect to see $\phi(\mathbf{x})$ controls for each treated individual at $\mathbf{x}$; so, with infinite data, we can match each treated individual to at most $\phi(\mathbf{x})$ controls, and even $\phi(\mathbf{x})$ controls is quite optimistic in a finite sample from that population. For instance, if $e(\mathbf{x}) = 0.4$, then $\phi(\mathbf{x}) = 0.6 / 0.4 = 1.5$, and pair matching may be the best we can hope to achieve at $\mathbf{x}$; however, if $e(\mathbf{x}') = 0.1$ then $\phi(\mathbf{x}') = 0.9 / 0.1 = 9$, and it should be possible to match with multiple controls at $\mathbf{x}'$.

2.2 Should We Focus on the Average Effect on the Treated?

If (1) were true and a sample of treated individuals were exactly pair matched with controls for $\mathbf{x}$, then the average pair difference is unbiased for the average effect $\mathrm{E}(r_T - r_C|Z = 1)$ of the treatment on the treated (Rosenbaum and Rubin, 1985). Write $\varrho(\mathbf{x}) = \mathrm{E}(r_T - r_C|\mathbf{x})$, which equals the estimable quantity $\mathrm{E}(r_T|Z = 1, \mathbf{x}) - \mathrm{E}(r_C|Z = 0, \mathbf{x})$ when (1) is true. Notably, if (1) held, then $\mathrm{E}(r_T - r_C|Z = 1)$ is not much affected by $e(\mathbf{x}) = \Pr(Z = 1|\mathbf{x})$ being low for $\mathbf{x} \in \mathcal{C}$, because $\mathrm{E}(r_T - r_C|Z = 1) = \mathrm{E}\{\varrho(\mathbf{x})|Z = 1\}$ equals

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$$\mathrm{E}\left\{\varrho(\mathbf{x})\middle|\mathbf{x}\notin\mathcal{C},Z=1\right\}\Pr\left(\mathbf{x}\notin\mathcal{C}\middle|Z=1\right)+\mathrm{E}\left\{\varrho(\mathbf{x})\middle|\mathbf{x}\in\mathcal{C},Z=1\right\}\Pr\left(\mathbf{x}\in\mathcal{C}\middle|Z=1\right)$$

where by Bayes theorem, $\Pr\left(\mathbf{x}\in\mathcal{C}\middle|Z=1\right)$ is not large because $e(\mathbf{x})=\Pr\left(Z=1\middle|\mathbf{x}\right)$ is not large for $\mathbf{x}\in\mathcal{C}$. If (1) were true, the parameter $\mathrm{E}\left(r_T-r_C\middle|Z=1\right)$ is not acutely dependent on the limited access to treated responses, r_T, for people with $\mathbf{x}\in\mathcal{C}$.

Consider an alternative estimate of $\mathrm{E}\left(r_T-r_C\middle|Z=1\right)$ obtained from I_1 treated-control matched pairs with $\mathbf{x}\notin\mathcal{C}$, and I_2 treated individuals matched to $K'>1$ controls for $\mathbf{x}\in\mathcal{C}$, where $I=I_1+I_2$, and then taking the unweighted average over *I* matched sets of the *I* treated-minus-average-control differences in outcomes within these *I* matched sets of unequal sizes. If (1) were true, this estimate remains unbiased for $\mathrm{E}\left(r_T-r_C\middle|Z=1\right)$ with no assumption that $\varrho(\mathbf{x})$ is constant. Does this alternative estimate have a much smaller standard error? Consider an admittedly too simple toy model, namely a linear block model with independent errors ε_{ij} and constant effect $\varrho(\mathbf{x})=\tau$:

$$R_{ij}=\beta_i+\tau Z_{ij}+\varepsilon_{ij} \text{ with } \mathrm{var}\left(\varepsilon_{ij}\right)=\varsigma^2. \quad (2)$$

Then the pairs-only estimate of $\mathrm{E}\left(r_T-r_C\middle|Z=1\right)$ has variance $2\varsigma^2/I$, while the blended estimate has variance $\left(I_1+I_2\right)^{-2}\left\{2I_1\varsigma^2+\left(1+\frac{1}{K'}\right)I_2\varsigma^2\right\}$. Under this toy model with $I=I_1+I_2=1212+213$ and $K'=4$, as in the example in §4, the blended estimate has a standard error that is 0.97 of the pairs-only estimate; so, the gain from $K'=4$ rather than $K'=1$ is quite small, essentially because I_2 is small compared to I_1.

There is more to this story: condition (1) is problematic in every observational study and in §4 it helps to increase K' when addressing this central issue. In §4, a plot facilitated by a transparent design shows model (2) is implausible in a scientifically interesting way.

2.3 Sensitivity Analysis and Design Sensitivity

Sensitivity to unmeasured bias in treatment assignment is appraised in term of parameter $\Gamma\geq 1$ that bounds the degree to which ignorable treatment assignment (1) may be violated for two people with the same $\mathbf{x}$ but different potential outcomes $\left(r_T,r_C\right)$ and $\left(r_T',r_C'\right)$:

$$\frac{1}{\Gamma}\leq\frac{\zeta\left(r_T,r_C,\mathbf{x}\right)\left\{1-\zeta\left(r_T',r_C',\mathbf{x}\right)\right\}}{\zeta\left(r_T',r_C',\mathbf{x}\right)\left\{1-\zeta\left(r_T,r_C,\mathbf{x}\right)\right\}}\leq\Gamma \text{ for all } \mathbf{x},\ \left(r_T,r_C\right) \text{ and } \left(r_T',r_C'\right). \quad (3)$$

Then $\Gamma=1$ implies (1), and one obtains randomization tests and confidence intervals for treatment effects (Lehmann, 1975). For $\Gamma>1$ one obtains an interval of *P*-values, an interval

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of point estimates, or an interval of endpoints for a confidence interval, where these intervals become wider as Γ increases (Rosenbaum, 2002, 2025).

If (1) were true, the investigator would not know it, and the best that could be said is that it would take a large bias in treatment assignment, Γ, to explain the data as something other than an effect caused by the treatment. There is a number, $\tilde{\Gamma}$, called the design sensitivity such that, when (1) holds and there is a specific treatment effect, the *P*-value testing the hypothesis of no effect tends to 0 as $I \to \infty$ for $\Gamma < \tilde{\Gamma}$, and it tends to 1 for $\Gamma > \tilde{\Gamma}$, so $\tilde{\Gamma}$ is the limiting sensitivity to unmeasured bias as the sample size increases; see Heng et al. (2021); Howard and Pimentel (2021) and Rosenbaum (2004, 2024, 2025).

To illustrate, compare the design sensitivity of two test statistics for *I* treated-minus-control matched pair differences, $Y_i = \left(Z_{i1} - Z_{i2}\right)\left(R_{i1} - R_{i2}\right)$, $i = 1, \ldots, I$ under model (2) with Normal errors, so $Y_i \sim N\left(\tau, \sigma^2\right)$ where $\sigma^2 = 2\varsigma^2$. Wilcoxon's signed rank statistic is (almost) a U-statistic that examines two pair differences at a time, Y_i and $Y_{i'}$, in all $\binom{I}{2}$ ways, and counts the number of times that the larger absolute difference is a positive difference, either $Y_i = |Y_i| > |Y_{i'}|$ or $Y_{i'} = |Y_{i'}| > |Y_i|$; see Lehmann (1975, p.368). Another U-statistic called U878 looks at 8 pair differences in all $\binom{I}{8}$ ways and counts the number of positive differences among the 2 of 8 differences with the largest $|Y_i|$; see Rosenbaum (2011). If $Y_i \sim N\left(\tau, \sigma^2\right)$ with $\tau / \sigma = 1/2$, then $\tilde{\Gamma} = 3.2$ for Wilcoxon's signed rank statistic and $\tilde{\Gamma} = 5.1$ for U878; so, in a sensitivity analysis conducted with $\Gamma = 4$, the power of this Wilcoxon statistic is tending to 0 as $I \to \infty$ and the power of U878 is tending to 1 Rosenbaum (2024, Table 1).

3 Considerations in Designing Observational Studies

3.1 Choose a Design to Increase Design Sensitivity

In an observational study, treatment and outcome may be associated in the absence of a treatment effect because of bias, measured or not, in who receives treatment and who receives control. Bias from measured covariates can often be removed by adjustments, such as matching, perhaps in combination with model-based adjustments (Rubin, 1979). Failure to control bias from a relevant unmeasured covariate produces a bias in the estimated treatment effect that is not diminished by increasing the sample size, *I*; so, it dominates the mean squared error of an estimated effect as $I \to \infty$, and does progressively more damage to the coverage rates of confidence intervals and the levels of tests.

Observational studies are therefore designed to increase design sensitivity, $\tilde{\Gamma}$, so that, after adjustments for measured covariates, an actual causal effect in the absence of unmeasured bias is likely to be found insensitive to small and moderate biases (Rosenbaum, 2004, 2020, 2025). That is, the design sensitivity anticipates the outcome of a sensitivity analysis that allows for unmeasured biases of various magnitudes, and the study is designed with the goal that it be insensitive to larger unmeasured biases, Γ.

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3.2 The Number of Controls in a Matched Set

Under simple, commonplace models — such as (2) — increasing the number of controls, *K*, in every matched set increases the design sensitivity, $\tilde{\Gamma}$. It also increases the Bahadur efficiency of a sensitivity analysis conducted with $\Gamma < \tilde{\Gamma}$, where efficiency drops to 0 as $\Gamma \to \tilde{\Gamma}$. In words, as *K* increases from 1 to 4, say, an actual treatment effect acquires an appearance that is somewhat more distinct from a bias in who is selected for treatment; consequently, a larger bias in treatment assignment, Γ, must be postulated to explain this now more distinct effect. This is fairly easy to prove (Rosenbaum, 2013, 2024), but it is also fairly intuitive. A positive treatment effect raises the response of the one treated individual in block *i* and leaves the controls unaffected. In the absence of a treatment effect, the responses in block *i* are from the same distribution, say $F_i(\cdot)$, and a positive selection bias departs from randomized treatment assignment by increasing the probability of assignment to treatment for individuals in block *i* with larger order statistics. In matched pairs, these two situations can look much the same. If there are $K = 4$ controls in blocks of size $J = K + 1 = 5$, the two situations look somewhat less similar, because the one treated individual has a response that stands a bit apart from the rest. Admittedly, for an effect of moderate size amid sampling noise, it is difficult to see this by eye, but appropriate test statistics can exploit this distinction between an effect without bias or bias without effect. For additional motivation and a detailed example, see (Rosenbaum, 2025, §10.2).

To illustrate, under the block model (2) with Normal errors, a design with $J = 5$ contributes $K = 4$ correlated treated-minus-control pair differences analogous to Y_i in §2.3. Where a signed-rank statistic for matched pairs, $J = 2$, scores the ranks of the absolute pair differences, a weighted rank statistic for blocks, $J \geq 2$, ranks the within-block ranges of responses and multiplies them by the within-block rank of the treated response (Quade, 1979; Rosenbaum, 2024; Tardif, 1987). If these *K* pair differences have expectation τ and variance $\sigma^2 = 2\varsigma^2$ with $\tau / \sigma = 1/2$, then the design sensitivity increases from $\tilde{\Gamma} = 3.2$ for Wilcoxon's signed rank statistic for pairs, $J = 2$, to $\tilde{\Gamma} = 4.8$ for the analogous statistic, Quade's statistic, for blocks of size $J = 5$; moreover, for the U-statistic U878, the design sensitivity increases from $\tilde{\Gamma} = 5.1$ for pairs with $J = 2$ to $\tilde{\Gamma} = 6.4$ with $J = 5$ (Rosenbaum, 2024, Table 1). In a sensitivity analysis at $\Gamma = 5.5$ using U878, as $I \to \infty$, the *P*-value tends to 0 with $J = 5$, but it tends to 1 with pairs, $J = 2$. A similar pattern of design sensitivities occurs with tests based on Huber's *M*-estimates, including the mean, and even if the block model (2) is replaced by a mixture effect (Rosenbaum, 2013, Table 5).

What happens using U878 with I_1 pairs and I_2 blocks of size $J = 5$? A natural but mistaken thought is that the design sensitivity must tend to a number between $\tilde{\Gamma} = 5.1$ and $\tilde{\Gamma} = 6.4$. Suppose that we combine the *P*-values for $J = 2$ and $J = 5$ using Fisher's method of combining independent *P*-values; then, the combination has the larger of the two design sensitivities, $\tilde{\Gamma} = 6.4$, because the *P*-value from the blocks with $J = 5$ is tending to 0 as $I \to \infty$ for all $\Gamma < 6.4$. For instance, *P*-values of 1 and 0.008 combine by Fisher's method to 0.047. A generalization of Fisher's method is Zaykin et al. (2002)'s truncated product method for combining independent *P*-values; see Hsu et al. (2013, §3) for a comparison with Fisher's method when used in sensitivity analyses.

3.3 Effect Modification and Design Sensitivity that Varies with **x**

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Effect modification means the treatment effect varies predictably as a function of the observed covariates. For instance, $\varrho(\mathbf{x}) = \mathrm{E}(r_T - r_C | \mathbf{x})$ might vary with $\mathbf{x}$. Other things being equal, a larger treatment effect is insensitive to larger biases; so, if there is effect modification then there may be a subpopulation defined by $\mathbf{x}$ in which the design sensitivity $\tilde{\Gamma}$ is larger (Hsu et al., 2013). Reasoning of this kind has often established that an ostensible treatment is insensitive to moderately large biases, Γ', in one subpopulation, thereby rendering more credible an ostensible effect insensitive to a smaller bias, $1 < \Gamma < \Gamma'$, in some other subpopulation. Moreover, based on considerations in §3.2, if we match K'-to-1 for $\mathbf{x} \in \mathcal{C}$ and in ratio K-to-1 with $1 \le K < K'$ for $\mathbf{x} \notin \mathcal{C}$, then we have reason to expect the design sensitivity to be larger for $\mathbf{x} \in \mathcal{C}$ even if $\varrho(\mathbf{x})$ is constant.

Obviously, $\varrho(\mathbf{x})$ might vary with $\mathbf{x}$, but this might not be evident by comparing $\mathbf{x} \in \mathcal{C}$ and $\mathbf{x} \notin \mathcal{C}$. Still, distinguishing $\mathbf{x} \in \mathcal{C}$ and $\mathbf{x} \notin \mathcal{C}$ is one natural place to look for effect modification, because people with $\mathbf{x} \in \mathcal{C}$ appear to be avoiding the treatment. The absence of effect modification can also be important. It suggests that the treatment might have a similar effect in other populations that have a different distribution of $\mathbf{x}$.

3.4 Counterfactual Risk and Design Sensitivity

Counterfactual risk is the risk of unfavorable outcomes in the absence of treatment. Suppose the outcome is nonnegative, $R \ge 0$, and large values of R are unfavorable. There is low counterfactual risk at $\mathbf{x}$ if $\mathrm{E}(r_C | \mathbf{x})$ is close to zero. In such settings, even if $\varrho(\mathbf{x}) = \varrho(\mathbf{x}')$ but $0 \doteq \mathrm{E}(r_C | \mathbf{x}) \ll \mathrm{E}(r_C | \mathbf{x}')$, the design sensitivity is often larger at $\mathbf{x}$ than at $\mathbf{x}'$, because the effect stands out more clearly (Rosenbaum and Small, 2017, §7.2). This is related to an older observation of Cornfield et al. (2009, p. 1186) that two subpopulations defined by $\mathbf{x}$ may have equal treated-minus-control risk differences, but one population may have a higher risk ratio due to a very low risk under control, and in this case there is less sensitivity to unmeasured confounding in the subpopulation with the higher risk ratio.

3.5 Evidence Factors

A study has two evidence factors if it permits two essentially independent tests of the hypothesis of no effect such that the unmeasured biases affecting each test leave the other test untouched (Karmakar, 2023; Rosenbaum, 2010, 2021; Zhao et al., 2022). Being essentially independent, two evidence factors may be combined by meta-analytic techniques as if they came from unrelated studies. Evidence factors are one formal view of the concept of triangulation in causal inference, in which two fallible sources of causal information provide mutual support be virtue of being susceptible only to very different unmeasured biases (Lawlor et al., 2016; Munafò and Davey Smith, 2018).

A second evidence factor is available in a matched design if the intensity of treatment for the treated individuals varies between matched sets (Rosenbaum, 2010, §5). The treated-minus-control difference in outcomes within I matched sets may exhibit an association with the I varying intensities of treatment between matched sets. If more intense treatment is expected to have larger effects than less intense treatment, then we might hypothesize that the treated-minus-control difference in outcomes will be larger in matched sets in which treatment

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occurred at higher intensity. This between-set information went untouched in the treatment-control comparison within matched sets in the first evidence factor.

With I_1 pairs and I_2 matched 1-to-K' sets, the treated-minus-mean-control outcome differences in the 1-to-K' sets are more stable. Under (2), a 1-to-K' difference has a variance that is $\left(1+\frac{1}{K'}\right)/2$ times the variance of a pair difference, or 0.625 for $K'=4$. Reduced variance can strengthen both evidence factors; see the Appendix.

3.6 A Form of Statistical Suffering

In 1992, the US National Academy of Sciences published a report assessing the quality of two studies — the "Longitudinal Study"and the "Immersion Study"— of the effectiveness of bilingual education in the US. The Academy was not entirely enthusiastic about the quality of these two studies, writing (Fienberg et al., 1992, p. 104):

Both the Longitudinal and Immersion Studies suffered from excessive attention to the use of elaborate statistical methods intended to overcome the shortcomings in the research design. Methods of statistical analysis cannot repair failures of conceptualization, design and implementation of studies … [An omitted list of methods of analysis]. The assumptions inherent in [these methods] at best finesse and at worst obfuscate issues that needed to have been dealt with explicitly in the study design and attendant analysis.

Helpfully, the Oxford English Dictionary offers several clarifying definitions of "obfuscate", including: "1. to cast into darkness or shadow; to cloud, obscure … 2b. to confuse, bewilder, or stupefy … 3b. to render something unclear, obscure, or confused." So, the quoted passage refers to several kinds of suffering.

Typically, controlled experiments are transparent in structure; so, the data look like what they are, with descriptive statistics, graphs, and formal analysis telling a single, consistent, accessible story. This is helpful too in observational studies — perhaps more helpful than in experiments, because genuine controversy attends many observational studies — but simple, transparent structure must accommodate adjustments for measured covariates. The use of matching as an aspect of adjustments often entails that formal analysis agrees with descriptive statistics and graphs of matched data. Introducing a felicitous phrase, Susser (1977, Chap. 7) describes matching as "simplifying the conditions of observation"with the goal of seeing more. Matching, even when combined with model-based adjustments, reduces dependence on modelling assumptions (Rubin, 1979).

Tukey (1980) stressed the importance of both confirmatory and exploratory analysis, and graphics are key to exploratory work. At times, designed research is conflated with confirmatory analysis, while immediately diving into data is conflated with exploratory analysis; however, careful design can support both confirmatory and exploratory analysis, and can sometimes support confirmation of planned explorations of effect modification in the specific sense in §3.3. The proposed design explores connections between outcomes, effects and a tendency to avoid the treatment, as illustrated in Table 2 of §4.2.

4 Example of the Design

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4.1 Matching Smokers to Controls in Two Ratios

Cigarette smoking is believed to cause some periodontal disease, meaning the separation of teeth and gums (Tomar and Asma, 2000). The 2009-2010, 2011–2012 and 2013-2014 US National Health and Nutrition Examination Surveys measured periodontal disease for individuals who were 30 years old or older. Here, the 1425 daily smokers in these surveys are compared to "never-smokers". A "never-smoker"smoked fewer than 100 cigarettes in their life, does not smoke now, and had no tobacco use of any kind in the past five days. Conversely, a daily smoker smoked at least 100 cigarettes in their life, currently smokes at least one cigarette every day, and did use tobacco in the past five days. Periodontal disease is measured by examining six locations on each of 28 teeth, if present, excluding wisdom teeth, recording the percent of locations exhibiting separation of gums and teeth.

The observed covariates $\mathbf{x}$ are female, age, income recorded as a ratio of family income to the poverty level, and education recorded as 1 for less than 9th grade, 2 for at least 9th grade but with no high school degree, 3 for a high school degree, 4 for some college or a two-year associate's degree, and 5 for at least a four year college degree. To protect confidentiality, NHANES caps reported age at 80 and income at 5. A propensity score was estimated using a linear logit model. In the US in 2009-2014, these covariate strongly predict smoking. The propensity score estimated that a women aged 80 with a college degree and an income of at least 5 times the poverty level had less than a 3% chance of smoking, whereas a man aged 33 with less than a 9th grade education and an income of 1/3 the poverty level had a 70% chance of smoking. At the covariates for that woman, we expect to find abundant nonsmoking controls for matching, so each such woman can be matched to several controls, but even 1-to-1 pair matching is challenging at the covariates for that man. Still, periodontal disease is rare at age 33 and more common at 80; so, graphs that describe the unmatched population are unlikely to resemble any analysis that takes account of covariates, such as age.

The 1212 daily smokers with propensity scores above 0.15 were matched 1-to-1 to 1212 never-smoking controls, and the 213 daily smokers with propensity scores of 0.15 or less were matched 1-to-4 to four to $852 = 4 \times 213$ never-smoking controls, as depicted in Figure 1(i). In Table 1 and Figure 1, it is seen that smokers and matched controls are similar to each other, but the 1-to-1 pairs look nothing like the 1-to-4 matched sets. The pairs are mostly men with limited education and income, while the 1-to-4 sets are mostly women with more education and income. In Table 1, the 90% quantile of income in the pairs is close to the 10% quantile of income in the 1-to-4 sets; so, the income distributions barely overlap. The upper quartile of education in the pairs equals the lower quartile of education in the 1-to-4 sets. The pairs are 39.6% female, but the 1-to-4 sets are 69.5% female. On average, the pairs are 7 years younger than the 1-to-4 sets.

The decision to match 1-to-1 above a propensity score of 0.15 and 1-to-4 below 0.15 was made by constructing several matched samples and checking for covariate balance. This choice was based on covariate balance alone, prior to examining any periodontal outcomes, so the decision is analogous to considering several experimental designs before choosing a good one and obtaining outcomes for that one design; see Rubin (2008). Much of the benefit of matching 1-to-K is achieved with $K = 4$; see Rosenbaum (2024).

4.2 Periodontal Disease Outcomes

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Figure 2 depicts cigarettes smoked per day and the periodontal disease outcome, namely the percent of diseased tooth locations. The number of cigarettes smoked turned out to be similar for the daily smokers in the 1-to-1 pairs and the 1-to-4 matched sets. This will be examined in §4.4. Not shown in Figure 2, being about 7 years older, the smokers in the 1-to-4 sets had smoked for about 5.4 more years than the smokers in the 1-to-1 pairs.

Periodontal disease is higher in smokers than in never-smokers. It is also higher among never-smokers in pairs than among never-smokers in 1-to-4 matched sets, and the same is true for smokers. Recall that income and education were typically higher in 1-to-4 sets than in pairs; so, it is conceivable that individuals in 1-to-4 sets had more access to dentists or engaged in more flossing of teeth or avoided other risk factors.

Table 2 is a sensitivity analysis for a one-sided test of the hypothesis of no effect of smoking on periodontal disease using the U878 statistic. Upper bounds on *P*-values are displayed for various $\Gamma \geq 1$, separately for 1-to-1 pairs and 1-to-4 matched sets, combining these two *P*-values using either Fisher's method or the truncated product method of Zaykin et al. (2002) truncating at 0.2. Notably, using the $I_1 = 1212$ pairs, rejection at level $\alpha = 0.05$ of no effect is insensitive to a bias of $\Gamma = 3$ but not $\Gamma = 4$, while rejection using $I_2 = 213$ matched 1-to-4 sets is insensitive to $\Gamma = 6$. A bias of $\Gamma = 3$ is a large departure from randomized treatment assignment, but it could not explain away the association between smoking and lung cancer in Hammond (1964)'s study; however, a bias of $\Gamma = 6$ could explain it (Rosenbaum, 2002, Table 4.2).

Use of either Fisher's method or the truncated product controls the family-wise error rate despite performing two tests, and if the two tests provide mutual support then their combined *P*-value bound can be smaller than both components, as does happen at $\Gamma = 4$. The truncated product (TP) reports insensitivity to a bias of $\Gamma = 5.5$ on the strength of the evidence provided by the 1-to-4 matched sets. Closed testing may be applied to the two component tests (Marcus et al., 1976).

How does this compare with a conventional design and analysis? For each of the $I_2 = 213$ blocks matched 1-to-4, one of 4 controls was selected at random, making $I = I_1 + I_2 = 1212 + 213 = 1425$ matched pairs at the bottom of Table 2. Using U878 as the test statistic, at $\Gamma = 4.2$, the *P*-value bound from $I = 1425$ pairs is 0.0779, similar to the $I_1 = 1212$ pairs with *P*-value bound 0.0745 at $\Gamma = 4$ in Table 2; whereas, the $I_2 = 213$ sets matched 1-to-4 have *P*-value bound 0.0476 at $\Gamma = 6$ in Table 2. If instead Wilcoxon's signed rank statistic is applied to all $I = 1425$ pairs, the upper bound on the *P*-value rounds to 1.0000 at $\Gamma = 4$, and is 0.35 at $\Gamma = 3$. In brief, a conventional paired design using Wilcoxon's signed rank statistic would report sensitivity to a bias of $\Gamma = 3$, while the fused design using U878 and the truncated product method reports insensitivity to $\Gamma = 5.5$ in Table 2.

Though few in number, the $I_2 = 213$ matched 1-to-4 sets substantially affect the degree of sensitivity to unmeasured bias. This was anticipated in §3.2 from theoretical calculations using block models (2). These 1-to-4 sets were available because the treatment, smoking, was rare for $\mathbf{x} \in \mathcal{C}$. However, it is clear the $I_1 + I_2 = 1425$ blocks in Figure 2(ii) were not generated by (2) because differences in shift, dispersion and shape are evident.

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4.3 Counterfactual Risk and Sensitivity to Unmeasured Bias

Although it is best to analyze periodontal disease (PD) as a continuous outcome, as in §4.2, insight is provided by considering the counterfactual risk of PD $\geq 20\%$ viewed as a binary outcome; see Table 3. As discussed in §3.4 and Rosenbaum and Small (2017, §7.2), theory suggests design sensitivity $\tilde{\Gamma}$ is often higher at an $\mathbf{x}$ where counterfactual risk is low. Notable in both Figure 2(ii) and Table 3, PD $\geq 20\%$ is rare among nonsmokers in the better educated, higher income 1-to-4 matched sets; it is 6.9%, rather than 17.7% for nonsmokers in the less educated, lower income matched pairs; i.e., counterfactual risk is lower in the 1-to-4 sets. This is particularly striking because the 1-to-4 sets are, on average, about 7 years older than the pairs, and periodontal disease increases with age.

The crude risk difference is similar in pairs and sets — $30.5-6.8=23.7\%$ for 1-to-4 sets and $39.9-17.9=22.0\%$ for pairs — but the Mantel-Haenszel odds ratio is larger for sets than pairs, 7.1 rather than 3.9. The Mantel-Haenszel odds ratio adjusts for matched covariates, including age. Viewed in terms PD $\geq 20\%$, there is little sign of effect modification in terms of the risk difference — 23.7% versus 22.0% — but there appears to be effect modification in terms of the odds ratio — 7.1 versus 3.9 — reflecting lower counterfactual risk among better educated, higher income nonsmokers.

4.4 Evidence Factors Using Cigarettes Per Day

In Figure 2, of the 1425 smokers, 267 smoked between 1 and 5 cigarettes per day, while 488 smoked at least 20 cigarettes per day. Perhaps, some people smoke compulsively all day long, while others smoke less because they are restricted during work hours, or smoke in specific social settings, or use cigarettes as an appetite suppressant.

The evidence factor analysis considers whether the number of cigarettes smoked by the smoker in Figure 2(i) predicts the treated-minus-control difference in periodontal disease within sets. For sets with $K' = 4$ controls, this is the difference between the treated outcome and the median of the four control outcomes. The association is depicted in Figure 3, where points in black are in the outer quartiles of the two variables, with the remaining points in gray; so, a cross has been cut from the scatterplot. The cross-cut odds ratio is based on counts of the corner points, and it is larger than one — more cigarettes are associated with a larger matched treated-control difference in periodontal disease — moreover, the odds ratio is larger for 1-to-4 sets at 4.67 than for 1-to-1 pairs at 1.99.

The cross-cut test uses the counts in the corners of Figure 3, and it generalizes the corner test of Olmstead and Tukey (1947), which cut at the median. The sensitivity analysis for the cross-cut test compares the corner counts to the extended hypergeometric distribution, and it has good design sensitivity: a bivariate Normal distribution with correlation 0.2 has $\tilde{\Gamma}=3.8$ for cuts at the quartiles (the default in the `crosscut` function of the `DOS2` package in `R`), or $\tilde{\Gamma}=5.0$ for cuts at the quintiles (Rosenbaum, 2016, Table 3). In pairs, rejection of no increase in effect with increased cigarette consumption has a *P*-value bound of 0.050 at $\Gamma=1.375$, whereas for 1-to-4 sets it has a *P*-value bound of 0.049 at $\Gamma=1.7$. In a $2\times2\times2$ table with two strata, for pairs and for 1-to-4 sets, the *P*-value bound is 0.049 at $\Gamma=1.59$.

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So, in yet another way, compared to the $I_1 = 1212$ pairs, the $I_2 = 213$ matched 1-to-4 sets have provided evidence that is more resistant to explanation as an unmeasured bias in the choice of smoking intensities by smokers.

5 Easy Steps To Building the Design

The matched design is constructed in two easy steps. First, the unmatched data are divided into two strata using the estimated propensity score, above κ or at most κ, where $\kappa = 0.15$ in §4. Then, a standard 1-to-*K* match is constructed for propensity scores above κ, and a standard 1-to-$K^{'}$ match is constructed for the remainder, where $K = 1 < 4 = K^{'}$ in §4. All details of matching in §4 are replicated in the documentation for `PeriUnmatched` in the `R` package `aamatch`. The choices $\kappa = 0.15$, $K = 1$ and $K^{'} = 4$ were based on several attempts to construct a well-balanced match. These choices were made prior to examining the periodontal disease outcome; so, they are part of the design of the study (Rubin, 2008). In §4, the two matches were: (i) exact for female and decade of age; (ii) applied fine balance to the propensity score, income and education, and (iii) used a robust Mahalanobis distance. The use of several matching techniques, (i)-(iii), by `aamatch`, together with checks on covariate balance in Figure 1 and Table 1, provide some protection against misspecification of the propensity score.

When graphing covariate balance in Figure 1 or outcomes in Figure 2, it is reasonable to separately plot 1-to-*K* and 1-to-$K^{'}$ sets, and other approaches may be reasonable also (Pimentel et al., 2015). If $K \neq K^{'}$, then it is not reasonable to plot all matched treated individuals in comparison with all matched controls, say in two parallel boxplots, treated and control. For example, the match with $K = 1 < 4 = K^{'}$ in §4 was exact for the covariate female; so, treated and control individuals with $K = 1$ were both 39.6% female, and treated and control individuals in 1-to-$K^{'}$ = 1-to-4 sets were both 69.5% female. However, the merged treated group is 44.1% female and the merged control group is 51.9% female.

6 Discussion

If there is a subpopulation $\mathcal{C}$ defined by measured covariates $\mathbf{x}$ such that treatment $Z = 1$ is uncommon for $\mathbf{x} \in \mathcal{C}$ but common for $\mathbf{x} \notin \mathcal{C}$, then the best possible matching ratio for $\mathbf{x} \notin \mathcal{C}$ may be low, say 1-to-*K*, but there is the opportunity to match at a higher ratio, 1-to-$K^{'}$ with $K^{'} > K$, for $\mathbf{x} \in \mathcal{C}$. Under simple block models (2), the increase from *K* to $K^{'}$ for $\mathbf{x} \in \mathcal{C}$ increases the design sensitivity, $\tilde{\Gamma}$, making the comparison insensitive to larger unmeasured biases in treatment assignment, Γ. Perhaps surprisingly, the design sensitivity $\tilde{\Gamma}$ for the entire study is not intermediate between the design sensitivity for its two components — the 1-to-*K* sets for $\mathbf{x} \notin \mathcal{C}$ and 1-to-$K^{'}$ sets for $\mathbf{x} \in \mathcal{C}$ — rather, it is the larger of the design sensitivities of the two components, as discussed in §3.2. If there are two evidence factors, both factors may benefit by increasing *K* to $K^{'}$ for $\mathbf{x} \in \mathcal{C}$.

When the block model (2) is false, the distinction between $\mathbf{x} \in \mathcal{C}$ and $\mathbf{x} \notin \mathcal{C}$ may still increase design sensitivity, perhaps by the increase from *K* to $K^{'}$ for $\mathbf{x} \in \mathcal{C}$, perhaps because of effect modification between $\mathbf{x} \in \mathcal{C}$ and $\mathbf{x} \notin \mathcal{C}$, or perhaps because of a difference in counterfactual risk in $\mathbf{x} \in \mathcal{C}$ and $\mathbf{x} \notin \mathcal{C}$. Finally, it is often of scientific interest to examine treatment effects for people who tend to avoid treatment.

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7 Appendix: Design Sensitivity and Evidence Factors

Section 3.5 related the treated-minus-mean-control difference in outcomes, say W_i, in set *i*, to the number of cigarettes, D_i, smoked by the smoker. In Figure 3, the cross-cut odds ratio is larger odds for $K' = 4$ controls than for $K = 1$ control. Is this expected?

Modify (2) to $R_{ij} = \beta_i + \delta D_i Z_{ij} + \varepsilon_{ij}$ where $\varepsilon_{ij} \sim_{\text{iid}} N\left(0, \varsigma^2\right)$, $D_i \sim_{\text{iid}} N\left(\mu_D, \varsigma_D^2\right)$ and Z_{ij} are independent with no bias in treatment assignment. With *K* controls, W_i has expectation $\delta\mu_D$, variance $\delta^2\varsigma_D^2 + \left(1 + \frac{1}{K}\right)\varsigma^2$, and D_i and W_i are bivariate Normal with correlation $\rho = \lambda / \sqrt{\lambda^2 + \left(1 + 1/K\right)}$ where $\lambda = \delta\varsigma_D / \varsigma$; so, ρ increases with *K*, with a consequent increase in design sensitivity $\tilde{\Gamma}$ for the cross-cut statistic (Rosenbaum, 2016, Table 3). For instance, cutting at the quartiles as in Figure 3, if $\lambda = 1/5$, then $\rho = 0.140$ for $K = 1$ and $\rho = 0.176$ for $K' = 4$; so, $\tilde{\Gamma}$ increases 2.5 for $K = 1$ to 3.2 for $K' = 4$.

8 Data Availability Statement

The data are in `PeriUnmatched` and `PeriMatched` in `R` package `aamatch` at CRAN.

SUPPLEMENTARY MATERIAL

In `aamatch` at CRAN, `PeriUnmatched` and `PeriMatched` replicate the match and analyses.

CONFLICT OF INTEREST

There are no financial or non-financial competing interests to report.

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Figure 1: Propensity score, education and income for $S_1 = 1212$ daily smokers in pairs, $N_1 = 1212$ never-smokers in pairs, $S_4 = 213$ daily smokers in 1-to-4 matched sets, $N_4 = 852 = 4 \times 213$ never-smokers in 1-to-4 matched sets. Means appear above boxplots.

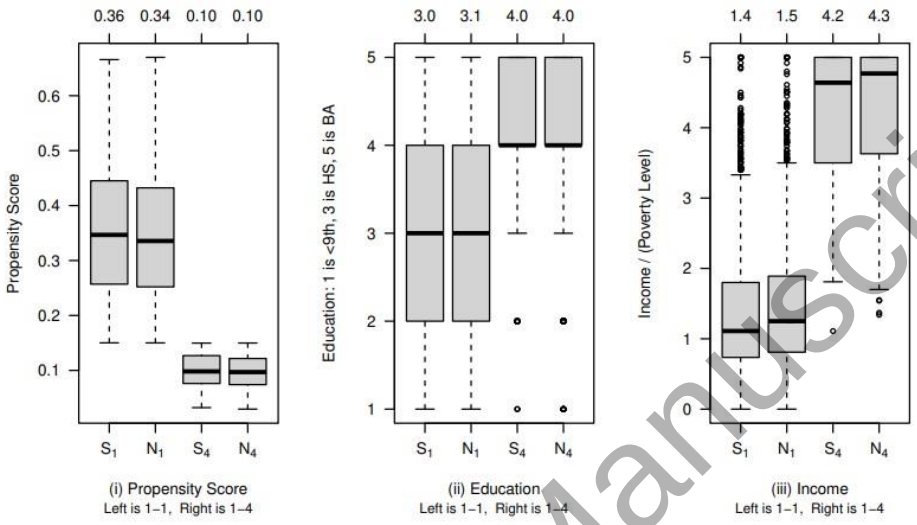

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Figure 2: Cigarettes smoked per day and periodontal disease for $S_1 = 1212$ daily smokers in pairs, $N_1 = 1212$ never-smokers in pairs, $S_4 = 213$ daily smokers in 1-to-4 matched sets, $N_4 = 852 = 4 \times 213$ never-smokers in 1-to-4 matched sets. Means appear above boxplots.

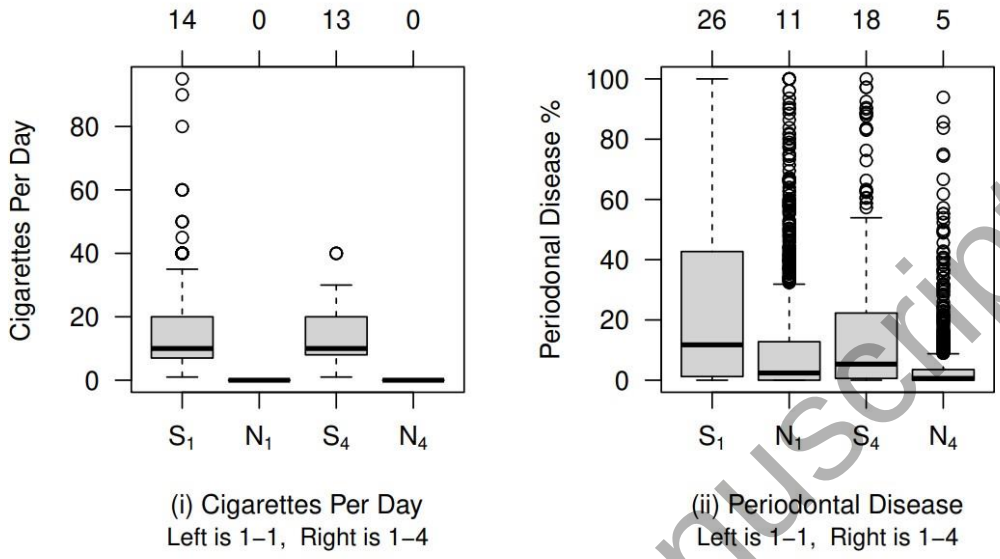

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Figure 3: Relationship between treated-minus-control difference in periodontal disease and cigarettes smoked per day by the smoker for $I_1 = 1212$ pairs and $I_2 = 213$ sets matched 1-to-4. Points in the outer quartiles of both variables are in black; others are in gray. The cross-cut odds ratio is based on counts of points in the corners.

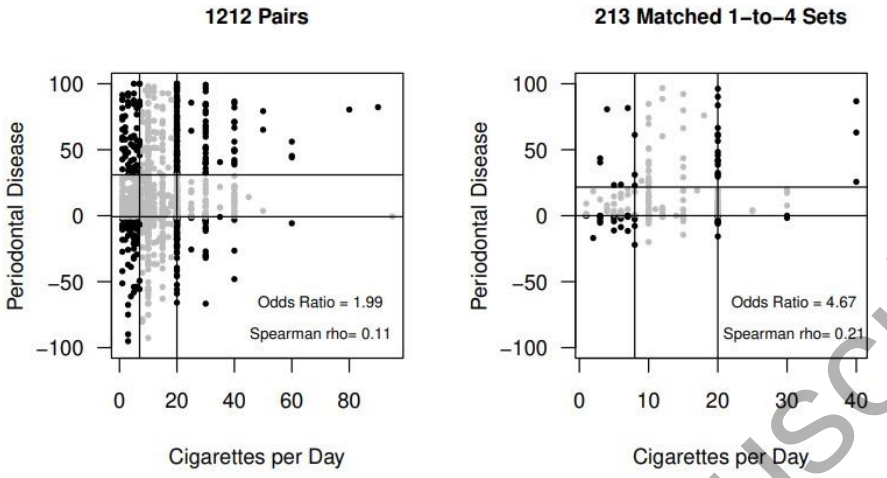

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Table 1: Covariates by group: female, age and income, education and the propensity score. S = daily smoker, N = never smoker. Income = ratio of family income to poverty threshold. Education is 1 for $<$9th grade, 3 for high school, 5 for $\geq$ college degree.

		Female	Age	Income			Education			Propensity		
	N	%	Mean	Mean	10%	90%	25%	50%	75%	Min	50%	Max
1212 Matched Smoker-Nonsmoker pairs with a higher propensity to smoke												
S	1212	39.6	46.6	1.4	0.3	2.9	2	3	4	0.15	0.35	0.67
N	1212	39.6	46.6	1.5	0.5	2.9	2	3	4	0.15	0.34	0.67
213 Matched Smoker-Nonsmoker matched 1-4 sets with a low propensity to smoke												
S	213	69.5	53.5	4.2	2.8	5.0	4	4	5	0.03	0.10	0.15
N	852	69.5	53.4	4.3	2.9	5.0	4	4	5	0.03	0.10	0.15

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Table 2: Sensitivity analysis showing the upper bound on the one-sided *P*-value testing H_0, using 1212 matched pairs, 213 sets matched 1-to-4, their combination using Fisher's method or the truncated product (TP), truncating at 0.2, or $1425 = 1212 + 213$ pairs. Bounds above 0.05 are in **bold**.

	Γ						
	1	2	3	4	5	5.5	6
Pairs 1-1	0.0000	0.0000	0.0001	**0.0745**	**0.6347**	**0.8660**	**0.9646**
Sets 1-4	0.0000	0.0000	0.0000	0.0008	0.0057	0.0117	0.0476
Fisher	0.0000	0.0000	0.0000	0.0006	0.0240	**0.0565**	**0.1873**
TP	0.0000	0.0000	0.0000	0.0005	0.0260	0.0447	**0.1161**
Comparisons: Analyses of just $1425 = 1212 + 213$ matched pairs							
U878	0.0000	0.0000	0.0000	0.0329	**0.5310**	**0.8148**	**0.9500**
Wilcoxon	0.0000	0.0000	**0.3549**	**0.9999**	**1.0000**	**1.0000**	**1.0000**

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Table 3: Counterfactual risk: Periodontal disease in $\geq 20\%$ of tooth locations in 1212 S-N pairs and 213 S-N 1-to-4 matched sets. The marginal odds ratio is computed from this table, while MH refers to the Mantel-Haenszel odds ratio that controls for 1212 pairs or 213 sets.

	1-to-1 Pairs		1-to-4 Sets	
	Smoker (S)	Never (N)	Smoker (S)	Never (S)
PD $\geq$ 20%	483	217	65	58
PD $<$ 20%	729	995	148	794
% with PD $\geq$ 20%	39.9	17.9	30.5	6.8
	Marginal	MH	Marginal	MH
Odds Ratio	3.0	3.9	6.0	7.1

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